# Calcurion Physics Worksheets

A-Level Physics

**Topic:** Magnetic Fields

**MARKSCHEME**

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**1.** $F = BIl\sin\theta$ **A1** [1]
**2.** $F = 0.15$ N **M1A1** [2]
**3.** Force, field, current mutually perpendicular **A2** [2]
**4.** $F = BQv\sin\theta$ **A1** [1]
**5.** $F = 3.2 \times 10^{-14}$ N **M1A1** [3]
**6.** Circle (or helix if component parallel) **A1** [1]
**7.** $BQv = mv^2/r$ **A3** [3]
**8.** $\Phi = BA\cos\theta$ **A1** [1]
**9.** $\Phi = 3.0 \times 10^{-3}$ Wb **M1A1** [2]
**10.** Induced e.m.f. proportional to rate of change of flux linkage **A2** [2]
**11.** Induced current opposes the change producing it **A2** [2]
**12.** $\mathcal{E} = 7.5$ V **M1A1** [3]
**13.** $V_p/V_s = n_p/n_s$ **A1** [1]
**14.** $V_s = 60$ V **M1A1** [2]
**15.** Increases flux linkage; concentrates field **A2** [2]
**16.** $v = BQr/m = 9.6 \times 10^6$ m/s **M1A1** [3]
**17.** $\sin\theta = 0$; $F = 0$ **A2** [2]
**18.** $\omega = 100\pi$; $\mathcal{E}_0 = NBA\omega = 15.7$ V **M1A1** [4]
**19.** Coil rotates; flux changes; e.m.f. induced; slip rings **A3** [3]
**20.** Output power = 60 W $<$ 100 W; not ideal / losses **A3** [3]
**21.** $B = \mu_0 I/(2\pi r) = 5.0 \times 10^{-5}$ T **M1A1** [3]
**22.** $F/l = \mu_0 I_1 I_2/(2\pi d) = 3.6 \times 10^{-5}$ N m$^{-1}$ **M1A1** [3]
**23.** Induced currents in core; laminate the core **A3** [3]
**24.** Place coil in field; withdraw quickly; measure charge/flux change **A3** [3]
**25.** $B$ from one wire; $F = BIl$ on other **A4** [4]